

IONIA COUNTY AP, MI, USA

WMO: 720661

Lat: 42.938N

Lon: 85.061W

Elev: 818

StdP: 14.27

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	7.6	-6.0	4.1	4.6	-0.3	5.6	10.2	26.8	28.1	24.5	29.4	5.2	290	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	90.2	71.7	87.6	70.8	84.4	69.5	75.1	84.7	73.5	82.8	72.2	80.8	10.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	123.4	79.6	71.1	118.1	77.9	69.7	112.4	76.7	39.0	85.4	37.7	82.7	36.4	80.4	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.4	18.2	-4.4	94.3	6.3	3.8	-8.9	97.0	-12.6	99.2	-16.1	101.3	-20.7	104.1
			-5.0	77.7	6.3	1.3	-9.5	78.6	-13.2	79.4	-16.7	80.2	-21.2	81.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	25.4	26.8	35.9	46.5	60.1	68.0	73.6	70.2	63.9	51.9	40.0	31.2
	DBStd	18.78	10.44	10.17	10.65	8.75	8.98	6.34	5.34	4.93	7.57	8.36	8.97	8.27
	HDD50	3036	762	651	458	166	16	0	0	0	1	80	320	582
	HDD65	6342	1227	1071	902	557	208	38	3	12	114	414	749	1047
	CDD50	2885	0	0	22	60	329	539	731	627	417	138	21	1
	CDD65	716	0	0	1	2	56	127	269	174	80	7	0	0
	CDH74	6440	0	0	13	40	581	1174	2598	1332	628	74	0	0
	CDH80	2012	0	0	1	3	173	354	947	343	184	7	0	0
Wind	WSAvg	8.2	9.5	9.4	8.9	9.8	8.5	7.6	6.1	5.7	6.6	8.3	9.2	8.8
Precipitation	PrecAvg	33.9	2.3	2.0	1.9	3.1	4.1	3.5	3.2	2.9	3.2	3.0	2.8	2.2
	PrecMax	41.4	4.6	4.6	3.9	6.3	9.8	7.6	5.5	6.5	8.8	6.0	6.9	5.4
	PrecMin	27.4	0.7	0.7	0.4	0.9	1.7	1.1	0.8	0.9	0.5	0.1	0.8	0.9
	PrecStd	4.4	1.2	0.9	1.0	1.3	2.2	1.8	1.3	1.5	1.9	1.7	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	58.9	77.4	79.4	89.7	91.9	96.9	90.3	90.3	81.0	70.4	58.8
		MCWB	53.2	50.6	61.7	60.9	68.8	72.6	73.5	71.5	69.8	63.4	58.1	54.3
	2%	DB	48.0	50.7	64.4	72.6	84.4	87.9	90.9	86.4	85.7	75.3	62.8	52.0
		MCWB	44.1	44.9	55.3	58.0	66.9	69.2	72.9	70.9	72.2	62.3	53.4	48.8
	5%	DB	41.4	44.3	58.5	66.2	81.1	84.1	88.1	83.8	81.0	70.9	58.7	46.3
		MCWB	38.0	39.7	51.2	53.7	65.0	68.3	71.7	70.0	68.2	61.9	51.8	44.0
	10%	DB	38.0	39.3	52.2	62.3	76.3	81.0	85.1	81.3	76.9	65.9	54.4	42.7
		MCWB	35.5	35.8	46.1	51.4	62.6	66.5	70.4	68.5	66.3	58.6	48.2	39.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	54.3	63.5	64.7	71.9	75.7	77.9	75.4	74.9	69.1	60.6	56.6
		MCDB	54.9	56.0	76.2	74.5	82.1	85.9	88.8	83.3	84.1	77.2	66.9	58.6
	2%	WB	44.4	45.9	58.2	59.4	69.4	73.4	75.7	73.8	73.0	65.8	55.9	48.8
		MCDB	46.9	51.0	63.5	67.5	80.8	83.0	86.3	82.3	81.5	72.4	62.0	51.0
	5%	WB	39.0	40.3	52.9	56.0	67.4	71.0	74.1	72.3	71.1	62.6	51.9	44.5
		MCDB	41.8	43.6	57.3	64.3	77.7	80.4	83.7	80.3	77.5	67.7	57.5	46.0
	10%	WB	36.2	36.3	46.0	52.5	64.8	68.9	72.6	70.7	68.8	59.6	48.7	39.8
		MCDB	38.0	39.5	50.9	60.6	73.2	77.3	81.7	77.6	74.7	66.7	53.9	42.1
Mean Daily Temperature Range	5% DB	MDBR	11.8	13.6	16.4	19.1	19.7	19.6	20.6	19.8	20.0	17.7	14.7	10.4
		MCDBR	16.5	17.3	24.3	27.0	24.4	22.7	23.3	22.3	22.8	24.8	21.3	16.0
		MCWBR	14.3	13.8	15.8	15.1	11.0	10.2	9.3	9.8	9.9	13.3	14.2	13.7
	5% WB	MCDBR	15.5	17.2	21.6	23.0	22.4	19.1	19.3	18.6	19.4	20.7	20.2	15.3
		MCWBR	13.8	14.7	15.8	15.7	11.4	10.7	9.3	9.6	9.8	14.5	15.4	14.5
Clear-Sky Solar Irradiance	taub	0.294	0.347	0.337	0.378	0.410	0.431	0.420	0.418	0.380	0.344	0.313	0.294	
	taud	2.349	2.168	2.365	2.276	2.236	2.215	2.249	2.253	2.345	2.445	2.477	2.406	
	Ebn at Noon	266	264	287	282	274	268	269	266	269	266	259	256	
	Edh at Noon	26	36	34	40	43	44	43	41	35	28	23	23	
All-Sky Solar Radiation	RadAvg	432	707	1130	1457	1704	1921	1949	1664	1337	840	512	349	
	RadStd	39	90	120	135	164	121	93	87	126	91	69	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page